

A Correlation-Aware Graph Transformer for Joint Regional Load and Operational Stress Forecasting in Bangladesh

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Abstract—Day-ahead grid planning needs both regional load forecasts and a view of system-wide operational pressure, yet these are often produced by separate tools. In Bangladesh, daily records from nine divisions report evening-peak load together with limitation, reserve, and shedding fields, which makes joint forecasting a natural fit. We build a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT) that reads a seven-day lookback with a train-estimated correlation graph and predicts regional demand and a bounded Operational Stress Index (OSI) from shared encoding and task-specific heads. On a held-out test year, correlation-graph configuration S2 gives the best joint results in our suite: macro demand MAE 88.65 MW and OSI R^2 0.745, with significant demand improvement over the historical hybrid reference (median Δ MAE -5.43 MW, $p = 5.5 \times 10^{-5}$). Ablations favour correlation adjacency over geographical or hybrid graphs and show a trade-off between demand-only accuracy and joint stress output. Post-hoc attribution highlights limitation and calendar coalitions, with only partial agreement across methods. All results are from offline chronological evaluation on Bangladesh division-level data; field deployment is left to future work.

Index Terms—Multi-task learning, spatio-temporal forecasting, operational stress index, correlation graph, smart grid, Bangladesh

I. INTRODUCTION

Grid operators need reliable day-ahead views of where load will rise and how tightly the system is constrained. In

developing economies, rapid demand growth often coincides with supply limits and load shedding, so planners cannot rely on demand forecasts alone; they also need a readable signal of operational pressure. Bangladesh’s national power system reports daily data for nine administrative divisions, with evening-peak demand appearing alongside limitation, reserve, and shedding fields in the same records [1]. Planning is therefore inherently spatial and system-wide at once.

In practice, however, load forecasting and stress monitoring are still commonly built as separate tools. Classical and ensemble models [2]–[4] predict demand without modelling links between divisions. Shallow recurrent or graph-convolution networks [5], [6] may struggle to combine a full weekly lookback with informative cross-node structure. Graph methods developed for residential or microgrid settings often assume geographical neighbours [7], which may not match how regional loads actually move together on a national footprint. Operational stress—driven by reserve shortfall, limitation pressure, and shedding—is rarely forecast jointly with regional demand; most shedding work [8] targets control rather than next-day stress indices. Explainability is also uncommon in graph-temporal models that output both load and stress [9], [10].

We treat day-ahead forecasting as a supervised multi-task problem [11] over nine divisions. Task 1 predicts next-day regional evening-peak demand in MW. Task 2 predicts a graph-level Operational Stress Index (OSI) in $[0, 1]$ that summarises shedding intensity, reserve margin, and limitation pressure. Demand and OSI are related in operations but not redundant: demand describes expected load by region, while OSI captures constraint pressure that demand alone does not fully explain. The question we ask is whether one spatio-temporal model can learn both outputs from historical Bangladesh grid records under strict chronological evaluation.

Our approach is the Correlation-Aware Multi-Task Forecasting Framework, implemented as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT). The model takes a seven-day lookback of division-level and national features together with a correlation graph estimated from training data, combines graph-aware spatial attention with temporal transformers [12], and decodes regional demand and bounded OSI through separate heads. Training uses a repaired multi-task loss so neither task is drowned out during optimisation. We evaluate against classical, recurrent, and graph-convolution baselines, run configuration ablations, apply Wilcoxon testing with Bonferroni correction, and report post-hoc attributions for the selected model.

This paper makes five contributions: (i) a correlation-graph PF-STGT for joint day-ahead demand and OSI forecasting on Bangladesh national data; (ii) evidence that correlation-only adjacency outperforms geographical and hybrid priors on this dataset; (iii) a practical multi-task loss calibration that stabilises joint training; (iv) a full benchmark and ablation programme with inferential testing; and (v) integrated explainability with multi-method attribution for both tasks.

Section II describes the method; Sections III–IV present experiments and results; Sections V–VI interpret the findings and conclude.

II. METHODOLOGY

A. Problem Formulation

Let $\mathcal{V} = \{1, \dots, N\}$ denote $N=9$ administrative divisions. At forecast origin t , the model receives a lookback window of $T=7$ days and predicts targets at horizon $H=1$. Node features $\mathbf{X}^{\text{node}} \in \mathbb{R}^{T \times N \times F_n}$ ($F_n=9$) encode division-level demand, supply, load, lags, rolling means, and regional shares. Global features $\mathbf{X}^{\text{global}} \in \mathbb{R}^{T \times F_g}$ ($F_g=17$) encode calendar indicators, grid aggregates, limitation variables, and national generation scalars. Task 1 outputs regional demand $\hat{\mathbf{y}}^{\text{demand}} \in \mathbb{R}^N$ (MW); Task 2 outputs graph-level OSI $\hat{y}^{\text{osi}} \in [0, 1]$.

OSI combines three train-normalised components:

$$\begin{aligned} c_1 &= L_{\text{total}}/D_{\text{total}}, \\ c_2 &= 1 - GR/\text{Highest_Gen}, \\ c_3 &= \text{TOL}/\text{Highest_Gen}, \\ \text{OSI} &= \text{mean}(\min_{\text{train}}(c_1), \min_{\text{train}}(c_2), \min_{\text{train}}(c_3)), \end{aligned} \quad (1)$$

where L_{total} , D_{total} , GR , and TOL are total shedding, demand, generation reserve, and operational limitation. Component c_1 tracks shedding relative to demand, c_2 captures reserve shortfall, and c_3 reflects limitation pressure. Regressing a continuous OSI preserves graded stress even on days without shedding. When forecasting $\text{OSI}(t+1)$, same-day OSI is excluded from inputs to avoid leakage.

The training objective pairs macro-averaged Huber loss on demand [13] with MSE on OSI:

$$\mathcal{L} = \mathcal{L}_{\text{demand}}^{\text{raw}}/100 + \lambda_2 \mathcal{L}_{\text{osi}}, \quad \lambda_2 = 20. \quad (2)$$

Raw demand loss is on the order of hundreds of megawatts, so without scaling it overwhelms the bounded OSI term and the model effectively trains as demand-only. Dividing by 100 MW and setting $\lambda_2=20$ brings both heads onto a comparable gradient scale.

B. Dataset and Graph Construction

We use the Bangladesh Smart Grid dataset (Mendeley Data, Version 5) [1]: 1,850 daily records from Nov. 2019 to Dec. 2024 across nine divisions. Chronological splitting yields 1,281 training, 263 validation, and 264 test sliding windows (Table I). Scaling, encoding, graph weights, and OSI bounds are fit on training data only.

Evening-peak demand across divisions moves together more through shared national growth and seasonality than through border adjacency. We therefore build a data-driven graph: pairwise Pearson correlations ρ_{ij} on training demand define undirected edges where $\rho_{ij} \geq 0.65$, weighted by ρ_{ij} and row-normalised to form **A**. This correlation graph keeps 33 of 36 possible pairs. Geographical adjacency (21 pairs) and hybrid adjacency (24 pairs) are retained only for ablation comparisons.

TABLE I
DATASET PROPERTIES AND S2 TRAINING CONFIGURATION.

Dataset	Value	Training	Value
Divisions (N)	9	Optimiser	Adam, $\text{lr } 5 \times 10^{-4}$
Window (T) / horizon	7 / 1 day	Batch / epochs	32 / ≤ 200
Node / global features	9 / 17	λ_2 (Eq. 2)	20.0
Train / val / test windows	1281 / 263 / 264	Parameters	749,058
Graph edges ($\tau=0.65$)	33 / 36 pairs	Early stop	Patience 15

C. PF-STGT Architecture

Fig. 1 shows the full pipeline; Fig. 2 details the selected S2 configuration. Node and global tensors pass through embedding layers with regional identity and positional encoding. A *spatial branch* applies graph-transformer self-attention over nodes at each timestep with additive bias from **A**:

$$\text{Attn}_{\text{spatial}}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B}_{\text{adj}}\right) \mathbf{V}. \quad (3)$$

A *temporal branch* runs a shared transformer encoder over each division's seven-day sequence. Learnable gating merges the two branches instead of fixing a spatial-then-temporal order, so the model can emphasise cross-division synchronisation

or within-division persistence as needed. The final timestep produces shared node embeddings $\mathbf{H}_{\text{shared}}$. A per-node linear head predicts regional demand; a graph-level head pools node states with global context and outputs OSI through a bounded activation. Both tasks share the encoder and differ only in the decoding layers.

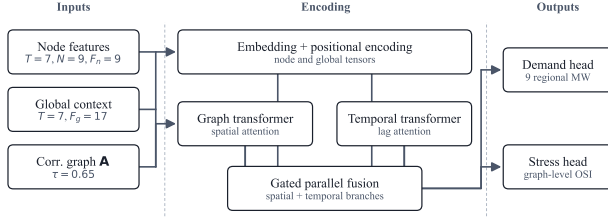


Fig. 1. End-to-end PF-STGT multi-task forecasting framework (frozen S2). Seven-day node and global tensors, a train-estimated correlation graph, parallel spatial-temporal encoding, and dual demand/OSI decoding heads.

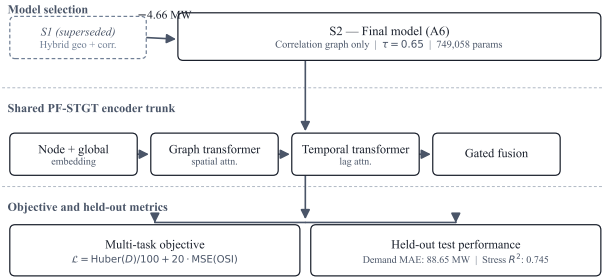


Fig. 2. Frozen S2 (A6) architecture with correlation-only graph ($\tau=0.65$). S2 replaces hybrid-graph S1 and reports held-out demand MAE of 88.65 MW and stress R^2 of 0.745.

III. EXPERIMENTAL SETUP

A. Baselines and Metrics

We compare the selected model S2 against seven references: linear regression, random forest [2], XGBoost [3], LSTM [5], GRU [14], T-GCN [6], and the historical PF-STGT reference B07 (hybrid geographical-correlation graph, W20 protocol). All models use the same chronological splits, feature store, windowing ($T=7, H=1$), and leakage rules, so differences reflect model design rather than preprocessing choices.

For Task 1 we report macro-averaged MAE (primary metric), RMSE, MAPE, and R^2 across nine divisions. Macro MAE is our main ranking criterion because dispatch planning cares about absolute megawatt error, not variance explained alone. Task 2 is scored with MAE and R^2 on OSI. We run paired Wilcoxon signed-rank tests [15] on per-window macro demand MAE with Bonferroni correction [16], and report bootstrap 95% confidence intervals [17] and Cohen’s d [18]. Hypothesis tests focus on demand MAE because that is the primary selection criterion; OSI results are reported as point estimates.

B. Training Configuration

S2 is trained with Adam [19] ($\eta=5 \times 10^{-4}$, weight decay 10^{-4}), batch size 32, up to 200 epochs with early stopping (patience 15 on composite validation score), and gradient clipping at 1.0. Model size is $d_{\text{model}}=128$, four heads, two spatial and two temporal layers (749,058 parameters). Deep baselines use AdamW for up to 60 epochs; classical models use training-set standardisation only. All runs use seed 42.

C. Ablation Design

Six PF-STGT configurations (A1–A6) isolate graph topology (none, geographical, hybrid, correlation-only), transformer removal, and single-task demand-only training. A1 matches B07; A6 matches S2. Together they show how much each design choice contributes to demand and OSI performance.

IV. RESULTS

A. Benchmark Performance

Table II and Fig. 3 report held-out test results ($n=264$ windows). S2 achieves the lowest macro demand MAE at 88.65 MW (RMSE 127.29 MW; MAPE 6.55%) and the best OSI R^2 at 0.745 (MAE 0.0371). Relative to B07, demand MAE improves by 4.66 MW (5.0%); relative to random forest B02, by 8.38 MW. Recurrent baselines B04 and B05 exceed 233 MW MAE with negative macro R^2 , and T-GCN B06 reaches 257.21 MW. Shallow graph convolution without full temporal-transformer capacity appears insufficient when nodes are densely correlated and the input window covers a full week. Classical ensembles still explain variance well—B02 reaches macro R^2 of 0.984—but they trail S2 on megawatt accuracy, which shows that R^2 and MAE can rank models differently when error concentrates in high-load divisions.

TABLE II
HELD-OUT TEST BENCHMARK COMPARISON ($n=264$ WINDOWS).

ID	Model	MAE	RMSE	MAPE	R^2	OSI MAE	OSI R^2
S2	Corr-Only PF-STGT	88.65	127.29	6.55	0.684	0.0371	0.745
B07	PF-STGT W20 hybrid	93.31	128.81	6.76	0.674	0.0499	0.585
B02	Random Forest	97.03	156.99	7.04	0.984	0.0481	0.555
B03	XGBoost	109.73	178.53	7.99	0.979	0.0497	0.525
B06	T-GCN	257.21	301.06	15.72	-0.483	0.0891	-0.304
Wilcoxon (demand MAE): S2 vs B07: median $\Delta = -5.43$ MW, $p = 5.5 \times 10^{-5}$, CI $[-7.17, -2.16]$							

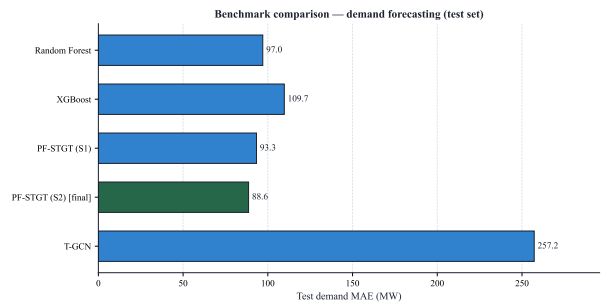


Fig. 3. Test-set macro demand MAE ($n=264$ windows). S2 achieves the lowest error (88.65 MW).

Wilcoxon tests on per-window macro demand MAE support the same ordering. S2 versus B07 gives median ΔMAE of -5.43 MW ($p=5.5\times 10^{-5}$) with bootstrap 95% CI $[-7.17, -2.16]$ MW. This is the only graph-variant comparison with statistically significant demand improvement over the historical hybrid reference. Using B07 as reference and Bonferroni $\alpha=0.0083$, B07 also beats B01–B06.

B. Regional and Stress Analysis

Per-region MAE under S2 ranges from 32.73 MW in Barishal to 293.98 MW in Dhaka. Dhaka accounts for about 35.7% of national demand and dominates the error budget; S2 still lowers Dhaka MAE by 5.80 MW versus B07, while peripheral divisions remain below 60 MW. Macro averages therefore understate how much capital-region accuracy matters for planning.

On OSI, S2 cuts MAE by 0.0128 and raises R^2 by 0.160 versus B07, explaining roughly three-quarters of held-out stress variance. Deep baselines B04–B06 return negative stress R^2 (-0.191 to -0.304), so their stress outputs are effectively uninformative. No paired Wilcoxon tests were run for OSI; stress claims rest on point estimates from the frozen test split.

C. Ablation Study

Table III and Fig. 4 compare configurations on the same test split. Correlation-graph A6 (S2) gives the lowest demand MAE among multi-task variants (88.65 MW) with OSI R^2 of 0.745. Geographical-graph A5 improves stress slightly (OSI MAE 0.0340, R^2 0.764) but costs 9.33 MW in demand error. Demand-only A4 reaches 86.89 MW without producing OSI—1.76 MW better than S2 on demand, which marks a real trade-off between pure load accuracy and joint stress output. Removing the transformer on the hybrid graph (A3) is not significantly different from A1 ($p=0.384$), but switching to a geographical graph (A5) significantly hurts demand ($p=1.48\times 10^{-4}$).

TABLE III
CONFIGURATION ABLATION ON THE HELD-OUT TEST SPLIT (A6 = S2).

ID	Variant	MAE	R^2	OSI MAE	OSI R^2
A4	Single-Task	86.89	0.731	—	—
A6 (S2)	Corr. Graph	88.65	0.684	0.0371	0.745
A3	No Transformer	92.64	0.671	0.0405	0.701
A1	PF-STGT W20	93.31	0.674	0.0499	0.585
A2	No Graph	93.93	0.701	0.0405	0.701
A5	Geo Graph	97.98	0.554	0.0340	0.764

D. Explainability

We apply post-hoc attribution [10], [20] to the frozen S2 checkpoint using grouped GradientSHAP, permutation importance, and spatial attention export, without retraining (Experiment 04). Fig. 5 shows mean spatial attention on the correlation graph, averaged over 24 case studies. Most mass falls on Dhaka (340.36), Rajshahi, and Khulna; peripheral divisions receive less. Attention correlates moderately with the frozen adjacency ($\rho=0.422$), so the learned spatial pattern partly follows the training-time correlation structure. Fig. 6

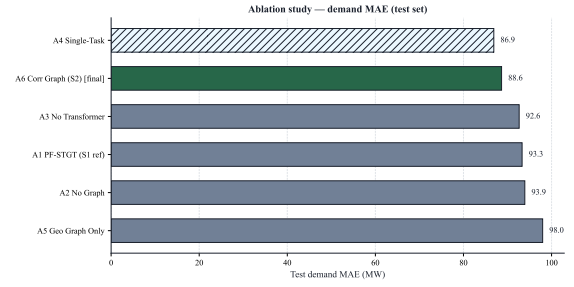


Fig. 4. Ablation demand MAE by configuration on the held-out test split. A4 is demand-only; S2 (A6) leads among multi-task variants.

plots grouped GradientSHAP magnitudes ($|\phi|$) for coalition registry G1–G11 (validation $n=20$). For stress, limitation stack (G8) and calendar/trend (G6) rank highest ($|\phi|=0.0191$ and 0.0190). For Dhaka demand, calendar (G6), engineered lags (G4), and national generation (G10) lead ($|\phi|=162.34, 101.26, 91.44$). The two tasks lean on different coalition families even though they share the same encoder.

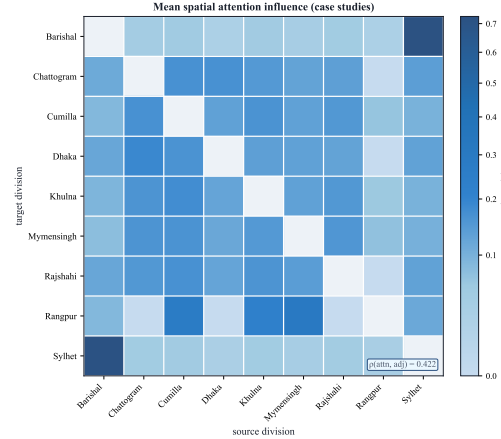


Fig. 5. Mean spatial attention influence on the correlation graph, averaged over 24 case studies. Cell (i, j) gives attention from source division j to target i ; diagonal entries are masked. $\rho(\text{attention}, \text{adjacency})=0.422$.

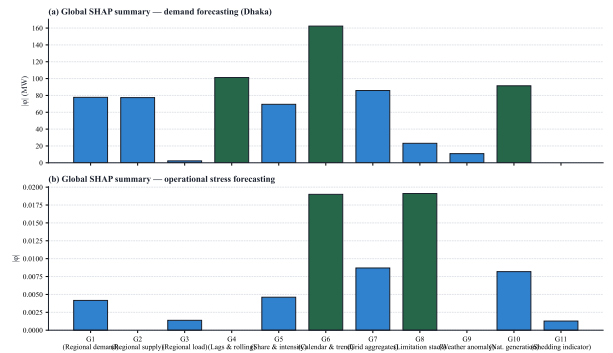


Fig. 6. Grouped GradientSHAP mean absolute attributions ($|\phi|$) on coalition registry G1–G11 (validation $n=20$). (a) Dhaka demand: calendar (G6), lags (G4), and national generation (G10) dominate. (b) Stress: limitation stack (G8) and calendar (G6) rank highest.

Comparing attribution methods, dual-path stress checks agree in 13 of 24 case studies (52.2%). Demand gradient and permutation rankings disagree ($\rho=-0.564$), while stress

rankings align more closely ($\rho=0.366$). These results describe how the model uses inputs on the evaluation sample; they should not be read as causal evidence.

V. DISCUSSION

The results show that one correlation-graph PF-STGT can produce coupled day-ahead load and stress forecasts from shared representations on Bangladesh national data. Below we discuss the main patterns and what they mean for this grid.

A. Correlation Graphs Versus Geography

Geographical adjacency is intuitive—divisions that share a border may exchange power—but it treats every border link equally regardless of how strongly loads actually co-move. On Bangladesh’s national footprint, evening-peak demand correlations are positive across divisions, driven by shared growth, holidays, and seasonality rather than border proximity. At $\tau=0.65$, the correlation graph keeps 33 of 36 pairs; geographical adjacency keeps only 21 border links and misses divisions that move together without sharing a border.

A5 (geographical graph) slightly improves OSI MAE (0.0340) but raises demand error by 9.33 MW over S2, which shows that the best graph for stress is not necessarily the best for load. Geographical conditioning significantly worsens demand ($p=1.48\times10^{-4}$) because it forces message passing through a sparse, equal-weight skeleton. Correlation weighting instead gives graded edges that match predictive coupling and can be refined further by graph attention (Eq. (3)). Similar behaviour has been reported on other national-scale systems where synchronisation dominates local exchange [21], [22].

B. Multi-Task Learning Trade-offs

Demand and OSI draw on the same calendar, limitation, and generation context but answer different questions: regional megawatt levels versus a single bounded stress score. Shared encoding lets the model pick up limitation-related structure—also visible in attribution for G8 (limitation stack) and G6 (calendar/trend)—that a demand-only model would have to recover indirectly. Without the loss rescaling in Eq. (2), demand gradients dominate and the OSI head barely trains.

Demand-only A4 reaches 86.89 MW, 1.76 MW below S2, because the stress head adds a mild regularising pull away from pure demand fitting. Joint training still delivers useful OSI output ($R^2=0.745$). In practice, a centre that needs only load forecasts may prefer A4; one that wants paired load and stress views can accept the small demand cost for much stronger stress predictions than B07 or classical baselines provide.

C. Bangladesh Grid Context

Three features of the dataset shape these findings. Nine divisions report shedding, reserve, and limitation in the same daily records, which makes composite stress supervision possible without extra telemetry. Dhaka dominates national demand ($\sim 35.7\%$) and error, so macro metrics understate how sensitive planning is to the capital region; that concentration appears across all model families and looks more like a

data-scale issue than a model failure. Finally, daily reporting with a seven-day lookback matches administrative practice, which helps explain why shallow recurrent and T-GCN models lag behind full transformer capacity on a densely connected correlation graph.

D. Deployment Notes

A single service could update regional evening-peak trajectories and a national OSI scalar from one seven-day input window each day. Operators might review both outputs during day-ahead planning, report Dhaka separately, and watch limitation-stack and calendar inputs flagged by attribution. We would expect shadow-mode deployment beside existing tools first, with division-level MAE checks and periodic stress-case review in place of formal prediction intervals. These are planning suggestions grounded in offline accuracy, not evidence of improved grid reliability or economic outcome in the field.

E. Limitations and Future Work

All evidence comes from one Bangladesh timeline at daily resolution with internal benchmarks only; there is no external replication or live deployment trial. Statistical testing covers demand MAE; OSI gains lack hypothesis tests or intervals. Attribution reflects model behaviour, not causation, and demand attributions disagree with permutation rankings ($\rho=-0.564$). Uncertainty quantification and full S2 residual diagnostics were not completed.

Natural next steps include repeating the protocol on comparable South Asian systems, testing adaptive correlation graphs, adding probabilistic heads, and running operator-in-the-loop studies of dual-task explanation dashboards.

VI. CONCLUSION

We asked whether one spatio-temporal model can jointly forecast next-day regional evening-peak demand and a graph-level Operational Stress Index from Bangladesh national grid records under chronological evaluation. The answer, on this dataset and protocol, is yes. Correlation-graph PF-STGT with parallel spatial and temporal encoding and repaired multi-task loss gives the best joint results in our benchmark suite: macro demand MAE 88.65 MW and OSI R^2 0.745, with statistically significant demand improvement over the historical hybrid reference.

Three findings stand out. Correlation adjacency works better than geographical priors here. Joint training trades a small amount of demand accuracy for substantially more useful stress output compared with demand-only training. Attribution analysis points to limitation and calendar coalitions for stress and to calendar, lag, and generation channels for Dhaka demand, with only partial agreement across methods.

The study adds a reproducible Bangladesh case for coupled load-and-stress forecasting and shows that graph design and loss weighting need empirical tuning rather than assumptions alone. A single model can support paired day-ahead forecasts when graph structure, multi-task calibration, and division-level monitoring are designed together—with the usual caveat that

our evidence is from offline evaluation, not from demonstrated field deployment.

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